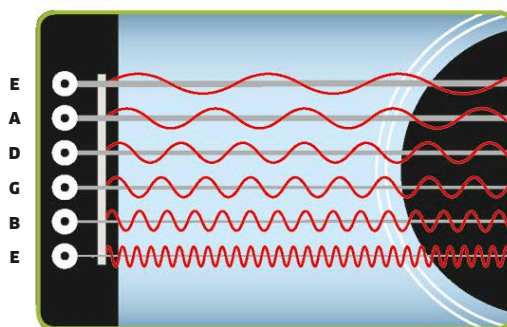


Acoustic guitar sounds

When a player plucks the strings of a guitar, each string vibrates at a different frequency to produce a note of a different pitch—higher- or lower-sounding. The pitch of the note produced depends on the length, tension, thickness, and density of the string. The strings' vibration passes into the body of the instrument, which causes air inside and outside it to vibrate, making a much louder sound.

**String thickness and density**

The thickness of strings affects their frequency and pitch: the thickest string creates the lowest frequency and lowest-pitch notes. Strings made of more dense materials will have a lower pitch.

Sound waves

Vibrating air comes out of the sound hole in waves that spread out evenly in all directions like the ripples in a pool of water.

Sound hole

Air at the sound hole oscillates, adding resonance.

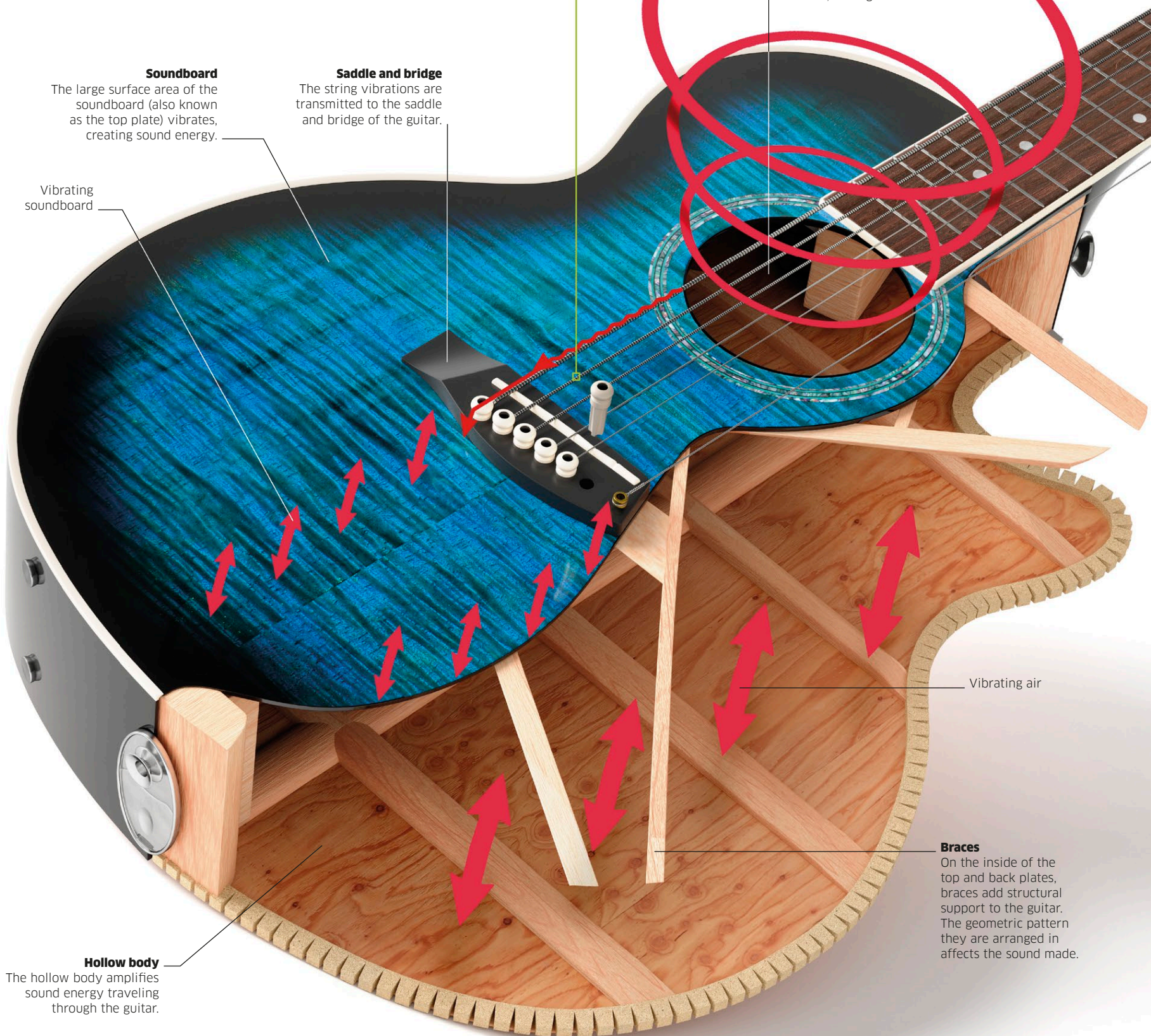
Soundboard

The large surface area of the soundboard (also known as the top plate) vibrates, creating sound energy.

Saddle and bridge

The string vibrations are transmitted to the saddle and bridge of the guitar.

Vibrating soundboard



Vibrating air

Braces

On the inside of the top and back plates, braces add structural support to the guitar. The geometric pattern they are arranged in affects the sound made.

Hollow body

The hollow body amplifies sound energy traveling through the guitar.